

Stage 1

Kernel
Telemetry
Capture

Stage 2

Event Parsing &
Feature
Encoding

Stage 3

Sliding-Window
Temporal Graph
Snapshot
Construction

Stage 4

GNN Structural
Embedding

Stage 5

Temporal
Transformer
Sequence
Encoding

Stage 6

Adaptive Risk
Scoring &
Classification

Stage 7

Interpretable
Alert
Generation

adaptive feedback: analyst-confirmed labels update graph and scoring thresholds